

MUHAMMAD HUSSAIN KHAN LODHI

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SUMMARY

Software engineer with a strong foundation in data structures, object-oriented design, and database systems, backed by shipped, tested code across the MERN stack, Java, and C/C++. Has designed multi-service systems, including a deterministic recommendation engine and a role-based access-control platform, with an emphasis on clean API design and code quality. BSCS graduate, CGPA 3.7/4.0.

EDUCATION

Sindh Madressatul Islam University Sep 2022 - Jun 2026
Bachelor of Science in Computer Science; CGPA: 3.7/4.0 Karachi, Pakistan
• **Relevant Coursework:** Data Structures, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks, Operating Systems, Mobile App Development

CERTIFICATIONS

IBM Full-Stack JavaScript Developer (Professional Certificate) Sep 2026
IBM, Coursera Online
Modern Web & App Development Mar 2026 - Present
Saylani Mass IT Training (SMIT) Karachi, Pakistan

EXPERIENCE

NextSkill Institute Jun 2026 - Sep 2026
Full Stack Development Trainee Karachi, Pakistan
• Completed a NAVTTC-affiliated software development training program with an IBM-aligned curriculum spanning front-end and back-end fundamentals
• Applied iterative, version-controlled development (Git/GitHub) across a progression of assignments of increasing complexity, from static UI to a multi-page e-commerce application
10Pearls - 10Shine Program Apr 2026 - May 2026
MERN Stack Development Intern Remote, Karachi
• Designed and implemented the data model and REST API for a full-stack Notes application (Node.js, Express, MySQL), including JWT authentication, email verification, and CRUD logic
• Authored unit/integration tests (Mocha, Chai, NYC coverage) and resolved issues flagged by SonarQube static analysis; collaborated via Git/GitHub with pull requests and code review
Self-Directed Learning & Development Sep 2022 - Present
Independent Study Karachi, Pakistan
• Applied data structures, OOP, and database design coursework to 10+ independently built projects spanning C/C++, Java, and the MERN stack
• Practiced collaborative software workflows (Git/GitHub) and iterative, test-driven development

PROJECTS

BuildVolt: PC-Build Recommendation SaaS (Final Year Project) GitHub | Live Demo
• Designed a deterministic compatibility engine for a PC-build recommendation SaaS, using regex-based spec inference (socket, RAM generation, wattage, form factor) to feed a budget-capped, PSU-safety-checked build assembler across 1,200+ lines of logic
• Built the REST API and data layer (Node.js, Express, Turso/libSQL) plus a WordPress/WooCommerce plugin (PHP), securing plugin endpoints with JWT auth, per-store rate limiting, and centralized secret management
Smart Complaint Management System GitHub | Live Demo
• Designed a role-based access-control system and a two-stage state machine, covering account approval and complaint lifecycle (Pending, In Progress, Resolved/Rejected), enforced across 15 REST endpoints
• Built the data model and API layer (Node.js, Express, MongoDB/Mongoose) with JWT auth, bcrypt hashing, and admin-only route guards; wrote a seed script to bootstrap the system securely
• Deployed backend and frontend as independent services on Vercel with environment-based CORS isolation
Student Records App GitHub | Live Demo
• Built a type-safe full-stack app (Next.js App Router, TypeScript, MongoDB/Mongoose) with a typed Student schema and REST-style API routes for full CRUD (list, create, update, delete)
• Built an admin dashboard surfacing live enrollment metrics (total, active, inactive, new-this-month) from the same typed data model, with search and end-to-end form validation

TECHNICAL SKILLS

Languages: Java, C++, C, JavaScript, TypeScript, SQL
Core CS: Data Structures, Object-Oriented Programming, REST API Design, Database Design, Unit Testing
Frameworks: React.js, Next.js, Node.js, Express.js
Databases: MySQL, MongoDB, SQL Server
Developer Tools: Git, GitHub, Postman, SonarQube, Mocha/Chai, VS Code, NetBeans
Deployment / Cloud: Vercel
Soft Skills: Problem-Solving, Teamwork, Communication, Adaptability, Time Management